

Kevin Wang

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EDUCATION

Stony Brook University

Stony Brook, NY

B.S., Chemistry, Concentration in Biological Chemistry

2020 – 2024

- Cumulative GPA: 3.94 / 4.00; *summa cum laude*
- Senior Thesis: “Viscosity of Unentangled Polymer Nanocomposites.”
- Advisor: Christian Aponte-Rivera

AWARDS AND HONORS

Chancellor’s Award for Student Excellence, SUNY	2024
American Chemical Society Division of Physical Chemistry Undergraduate Award, ACS	2024
Phi Beta Kappa, Stony Brook Chapter	2024
Chemistry Award for Outstanding Academic Achievement, Stony Brook University	2024
Presidential Scholarship, Stony Brook University	2020 – 24
National Merit Scholarship, Stony Brook University	2020 – 24
Dean’s List, Stony Brook University	2020 – 24
Outstanding Academic Achievement Award, Stony Brook University	2021, 2022, 2023
Upstander Award, Stony Brook University	2022

RESEARCH EXPERIENCE

Technician B

Jun 2025 – Present

Columbia University Irving Medical Center

New York, NY

- Conducted deep mutational scan of hepatitis B virus X protein (HBx) to assess functional effects of sequence variation
- Supported SARS-CoV-2 antiviral drug-resistance studies including execution and data collection
- Designed and cloned plasmids for cell-based assays across HBV, SARS-CoV-2, and RABV projects
- **Advisor:** Sho Iketani

Research Assistant

Feb 2023 – Jun 2025

Stony Brook University

Stony Brook, NY

- Presented senior thesis on quantifying the Einstein viscosity of unentangled polymer nanocomposites
- Conducted LAMMPS molecular dynamics simulations of polymer nanocomposites using the SeaWulf high-performance computing cluster
- Developed Python and R scripts to analyze simulation data, interpret results, and generate visualizations
- Created and implemented Python scripts to automate task submission and scheduling using Slurm
- **Advisor:** Christian Aponte-Rivera

NSF REU Research Fellow

Jun 2023 – Aug 2023

Worcester Polytechnic Institute

Worcester, MA

- Modeled Cu(I)-protein systems using a coarse-grained lattice-based model to predict binding energy
- Improved the parameters used to predict binding energy, resulting in a 41% increase in accuracy
- Developed Python and Unix shell scripts to automate program workflow and to process data
- Prepared and presented a research poster at the WPI Summer Research Showcase
- **Advisor:** George A. Kaminski

PUBLICATIONS

1. Chithelen, J., Lovett, D.H., **Wang, K.**, Torres Yanagisawa, A.E., Caccin, F., Iketani, S., 2026. Cross-resistance patterns in SARS-CoV-2 against 3CL protease inhibitors. *Nature Communications*. (accepted in principle)

PRESENTATIONS

1. “Viscosity of Unentangled Polymer Nanocomposites,” Senior Research Symposium, Stony Brook, New York, May 2024
2. “Predicting Cu(I) Binding Affinities Using a Coarse-Grained Lattice Model,” WPI Summer Research Showcase, Worcester, Massachusetts, August 2023

CLINICAL EXPERIENCE

Mental/Behavioral Health Clinic Intern

Jul 2022 – Aug 2022

NYC Health + Hospitals / Queens

New York, NY

- Collaborated with a team of interns and clinic staff in leading small group activity therapy for pediatric patients, teaching mindfulness and stress-reduction exercises
- Designed educational posters and newsletters for use in activity therapy and to promote the summer LEAD program
- Tutored 2 pediatric groups within the Behavioral Health Center and prepared them for the public-school curriculum

Medical Assistant

Dec 2021 – Jan 2022

Milestones Pediatrics of New York

New York, NY

- Greeted patients, checked them in and out of the clinic, and collected personal and medical information
- Assisted with patient care, including taking vital signs and explaining treatment procedures
- Maintained electronic medical records and scheduled appointments

TEACHING

Laboratory Design Teaching Assistant

Jan 2022 – May 2022

Stony Brook University Biology Department

Stony Brook, New York

- Assisted faculty in developing new experiments and refining existing experiments for future laboratory courses
- Coordinated with fellow teaching assistants to identify and resolve areas of confusion in experimental procedures
- Performed pilot testing of lab activities to validate protocols, streamline workflow, and ensure feasibility within time constraints
- Edited instructional materials for students on online learning management system to improve clarity

COMMUNITY INVOLVEMENT

Second Deputy Chief of Operations

Sep 2021 – May 2024

Stony Brook Campus Community Emergency Response Team

Stony Brook, New York

- Led a volunteer emergency response team of 80 members, overseeing training, deployment coordination, and inter-agency collaboration
- Conducted simulation drills and trained volunteers in basic medical aid, fire suppression, search and rescue, and triage in accordance with FEMA guidelines
- Oversaw captain recruitment, onboarding, and retention efforts, increasing captain capacity by 100% over a year

- Provided on-the-ground leadership during deployments, ensuring timely, organized action

Program Success Manager

Apr 2021 – Sep 2021

The City Tutors

New York, NY

- Managed a team of interns to oversee the operations of a CUNY-wide mentorship program
- Matched over 250 students with career development mentors based on student profiles and career interests
- Collected, analyzed, and synthesized survey data to evaluate program outcomes, identify process inefficiencies, and implement improvements

PROFESSIONAL MEMBERSHIPS

American Chemical Society

SKILLS

Programming Languages: Python, R, Unix/Linux, Mathematica

Software: ChemDraw, ImageJ, SnapGene, Prism, EndNote, MobaXterm, LAMMPS, VMD

Laboratory: Spectroscopy (UV/Vis, IR, NMR, MS), Chromatography (GC, HPLC, TLC), Synthesis, Purification, Mammalian Culture, Transfection, Transduction, qPCR, RT-PCR, Flow Cytometry, FACS, Gel Electrophoresis, DNA/RNA isolation, Confocal Microscopy

Certifications: Basic Life Support Certification (Exp. 2026), Narcan Trained

Languages: Fluent in English, conversant in Mandarin and Spanish (incl. Intermediate Medical Spanish)